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Therapeutic Monosaccharides: Looking Back, Moving Forward

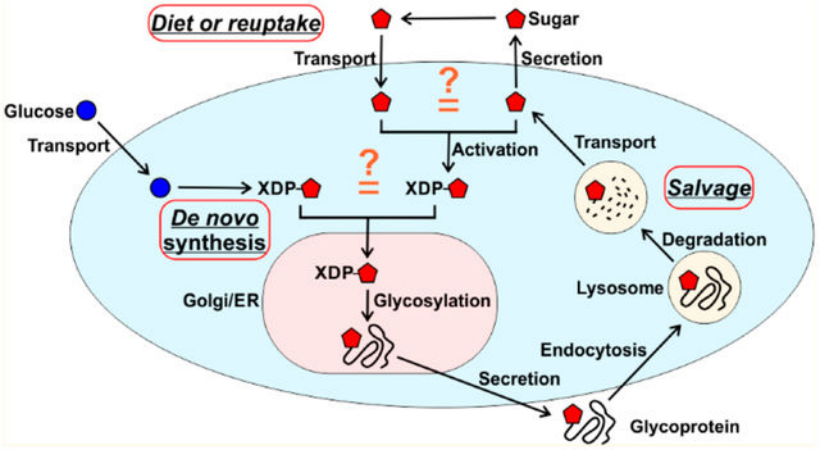
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Abstract

In this review, we focus on the metabolism of mammalian glycan-associated monosaccharides, where the vast majority of our current knowledge comes from research done during the 1960s and 1970s. Most monosaccharides enter the cell using distinct, often tissue specific transporters from the SLC2A family. If not catabolized, these monosaccharides can be activated to donor nucleotide sugars and used for glycan synthesis. Apart from exogenous and dietary sources, all monosaccharides and their associated nucleotide sugars can be synthesized *de novo*, using mostly glucose to produce all nine nucleotide sugars present in human cells. Today, monosaccharides are used as treatment options for a small number of rare genetic disorders and even some common conditions. Here, we cover therapeutic applications of these sugars and highlight biochemical gaps that must be revisited as we go forward.

Graphical Abstract



Far and away the most important role of monosaccharides is to provide energy for cells. Besides energy-generating catabolism, they can be utilized for the biosynthesis of multiple cellular components such as polysaccharides (e.g., glycogen and hyaluronan) and glycoproteins, proteoglycans, glycosphingolipids, and GPI anchors, all gathered under the

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biosynthetic umbrella of glycosylation. In mammalian cells, this enzymatic process involves at least nine monosaccharides (glucose, galactose, *N*-acetylglucosamine, *N*-acetylgalactosamine, xylose, glucuronic acid, sialic acid, mannose, and fucose) that are linked to proteins and lipids to generate hundreds of glycan structures (Figure 1).[1] Glycosylation is important for the biosynthesis, folding, stability, trafficking, and turnover of many different macromolecules, all of which play an essential role in the interactions between the cells, tissue morpho-genesis, and the immune response.[2]

In the diet, glucose is the most important monosaccharide. The other sugars occur in various polysaccharides that are mostly indigestible and not bioavailable but are often degraded by gut bacteria, producing short chain fatty acids, some of which can be absorbed by the host.[3] Besides their dietary origin, monosaccharides can be salvaged from glycans degraded in the lysosomes. They can be recycled within the cell or secreted.[1] The bulk of glycan monosaccharides is likely synthesized *de novo*, ultimately from glucose (Figure 1). This review will cover time-honored, well-established tenants of monosaccharide use in glycan synthesis. We will also show where gaps in our understanding exist, especially in the relative contributions of multiple pathways. There is an urgency here because many simple sugars are serving as therapeutics for rare diseases and more common conditions.

BASICS OF GLYCAN-ASSOCIATED MONOSACCHARIDE BIOCHEMISTRY

Incorporation of monosaccharides into glycans first requires activation by the ATP-dependent specific kinases. These activated monosaccharide phosphates require further conversion to nucleotide sugars using UTP-, GTP-, or CTP-dependent reactions that generate free pyrophosphate and the activated pyrophosphate-containing sugar donors (Figure 1) Most nucleotide sugars are synthesized in the cytosol,[4] except for CMP-sialic acid, which is made in the nucleus.[5] Because glycosylation occurs predominantly in the Golgi apparatus and the endoplasmic reticulum (ER),[1] the donors must be delivered there using the appropriate nucleotide sugar transporters (antiporters) (Figure 2).[6] In the organellar lumen, the co-localized donors and respective glycosyltransferases add the monosaccharide to the acceptors, initiating glycan synthesis and providing glycans for further extension.[1] An exception to this transport rule is UDP-*N*-acetylglucosamine (UDP-GlcNAc), which is utilized in the cytosol for protein modification by O-GlcNAc transferase or hyaluronan biosynthesis that requires cytoplasmic UDP-GlcNAc and UDP-glucuronic acid.[7,8]

DIETARY CARBOHYDRATES AFFECT THE HOST GUT MICROBIOME

Carbohydrates are important compounds in an infants' diet. In addition to lactose, human milk contains more than 200 unique oligosaccharides.[9] They occur both as free glycans and as glycoconjugates.[10] The precise composition of milk glycans varies among women and during lactation.[11] In contrast to human milk, infant formulas have a more limited amount and a more limited variety of free oligosaccharides.[12] Interestingly, infants are already exposed to milk oligosaccharides in the uterus as they are found in amniotic fluid, highlighting their importance.[13]

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Besides oligosaccharides, human milk contains different species of bacteria, including *Bifidobacterium*, *Bacteroides*, and *Ruminococcus*, which help in oligosaccharide digestion and assimilation qualifying milk as a probiotic food.[14,15] Oligosaccharides present in human milk are an important factor that help to shape the gut microbiota and health of infants.[16] Synergistic actions of sialylated milk oligosaccharides and infant intestinal microflora promote growth and have an impact on metabolism in different organs.[17] Unlike breast-fed infants, newborn babies fed with formula develop different gut microbiota, with a lower content of beneficial bacteria.[18]

Modulating the host gut microbiome with dietary-provided mono- and oligosaccharides may improve the health and quality of life. However, at this point, we do not know the extent to which complex oligosaccharides directly contribute to host glycan synthesis versus serving as microbiome modulators.

CONGENITAL DISORDERS OF GLYCOSYLATION

Pathogenic mutations in glycosylation-related genes can perturb cellular glycosylation, resulting in developmental and physiological abnormalities in many organ systems.[19] Congenital disorders of glycosylation (CDG) make up a rapidly expanding group of more than 135 rare metabolic diseases.[20] They are rare because glycans are critical for development and many mutations in glycan pathways are probably lethal. More than half of the identified CDG types are due to deficiencies in the N-glycosylation pathway.[20] Most of these disorders can be initially identified by carbohydrate-deficient transferrin (CDT) analysis, examining glycosylation of this abundant serum protein.[21] However, in some CDGs, abnormal transferrin glycosylation is not seen or normalizes over time in authentic patients. New biomarkers are required to improve the diagnostic process, but ultimately, genetic analysis is necessary.[22]

Although the number of identified disorders is growing, there are few treatment options. Several therapeutic strategies rely on supplementing the individual's diet with the limiting monosaccharides.[20] Because there are few data on the relative contribution of each monosaccharide from diet/salvage to glycan production, it is impossible to predict which rate-limiting metabolic steps might respond to modest increases in circulating levels of monosaccharides. This is especially important for individuals with impaired glycosylation.

D-MANNOSE

D-Mannose is required for N- and O-glycosylation, C-mannosylation, and GPI anchor synthesis. The mannose concentration in the human plasma is ~50 μM, which primarily comes from N-glycan processing.[23] Members of the SLC2A family of hexose transporters commonly termed "GLUT" carry nearly all monosaccharides, including mannose, across the plasma membrane into the cytoplasm.[23] Inside the cell, mannose is phosphorylated by hexokinase (HK) to produce mannose 6-phosphate (Man-6P),[24] which can further serve as a substrate for three different pathways (Figure 1). First, it can be converted to mannose 1-phosphate (Man-1P) and further to GDP-mannose (Figure 1)[25] and subsequently to dolichol-phosphate mannose, both of which can be directly utilized for N-glycan biosynthesis.

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Alternatively, Man-6P can serve as a substrate in 2-keto-3-deoxy-D-glycero-D-galactonononic acid (KDN) synthesis (Figure 1). The function of this pathway is unknown, but providing a high mannose concentration to cells increases the production of KDN.[26] Man-6P can also be isomerized to fructose 6-phosphate (Fruc-6P), allowing it to be catabolized through glycolysis or to serve as a substrate for the production of different nucleotide sugars (Figure 1).[27] While mannose is the most abundant monosaccharide in N-glycans, <2% of the mannose entering the cell is utilized for glycosylation. The great majority is converted to Fruc-6P and catabolized, presumably through glycolysis.[23] Finally, GDP-mannose can be converted to GDP-fucose, resulting in its incorporation into glycans as fucose (Figure 1).[28]

Both mannose and glucose contribute to GDP-mannose synthesis through Man-6P (Figure 1);[29] however, cells labeled with differentially tagged ^{13}C mannose and glucose showed that under physiological conditions (5 mM glucose and 50 μM mannose), glucose is the major source (60–80%) of N-glycan mannose.[30] Under this condition, ^{13}C mannose cannot be detected in glycogen, galactose, or GlcNAc in N-glycans. However, when its concentration is increased to 1 mM and it is the sole monosaccharide source of mannose in N-glycans, mannose can be seen to contribute to glycolysis, galactose and GlcNAc, but it is never found in glycogen,[30] implying some segregation of metabolic pathways. Mannose taken up by the cells is handled differently than after it is released within the cells due to N-glycan processing. Mannose released by intracellular processing is secreted from the cells as free mannose, suggesting that the monosaccharide origin is important for its future destination.[23]

Glycan-associated metabolism of mannose is significantly perturbed in at least two CDGs caused by mutations in phosphomannose isomerase (*MPI*) and phosphomannomutase 2 (*PMM2*) (MPI-CDG and PMM2-CDG, respectively). PMM2-CDG is the most prevalent CDG type, with more than 900 reported cases.[31] It is caused by pathogenic variants in *PMM2*, whose encoded protein converts Man-6P to Man-1P, an intermediate in GDP-mannose synthesis (Figure 1). Reduced PMM2 enzymatic activity ultimately leads to depleted incorporation of mannose into glycoproteins and truncation of the lipid-linked oligosaccharides under low-glucose conditions.[29] Some of these truncated glycans can still be transferred to proteins, as shown by recent analysis of patient sera.[22] In MPI-CDG, the level of conversion of Fruc-6P to Man-6P is decreased because of reduced MPI activity (Figure 1), resulting in an abnormal serum glycosylation pattern. MPI-CDG was the first CDG type found to be effectively treated by dietary supplementation with a monosaccharide.[32] Affected individuals responded quickly to mannose therapy, with correction of the glycosylation defect. However, liver abnormalities in some patients were irreversible and progressed to liver fibrosis that required a transplant.[33]

Mpi hypomorphic mice carrying a known pathogenic variant that generated a patient level of enzymatic activity showed very few abnormalities, unless mice were stressed.[34] However, supplementing dams drinking water with 1–2% mannose reduced litter size and pup survival in a dose-dependent manner. Moreover, the increased level of mannose and reduced Mpi activity resulted in highly localized eye defects due to the failure of normal lens development along with the accumulation of Man-6P and related metabolites.[34] Considering the data from *Mpi* hypomorphic mice, pregnant, at-risk women should be wary of using

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mannose as a preemptive therapy, because it could be harmful to a genetically compromised fetus. Besides being used to treat MPI-CDG patients, mannose has been extensively used as both a treatment and a prophylactic for urinary tract infections because its excretion in the urine competes with lectin-mediated binding of *Escherichia coli* to epithelial cells.[35]

In more recent studies, mannose showed therapeutic promise in other areas such as obesity and cancer. For example, when mannose was provided to mice early in life, it prevented high-fat-diet-induced obesity and promoted the lean and fit phenotype that correlated with changes in the gut microbiota.[36] Furthermore, mannose was shown to cause the arresting of growth in several tumor types in mice as well as increased tumor cell death upon chemotherapy, suggesting mannose as a selective therapy for the treatment of cancer.[37] While the mechanism has not been demonstrated, it is most likely due to the toxic accumulation of Man-6P that impairs glycolysis, starving the cells of metabolic fuel.

L-FUCOSE

L-Fucose is incorporated into N- and O-glycans, and glycolipids, using GDP-fucose as the donor substrate.[38] GDP-fucose also can be synthesized *de novo* from either glucose or mannose. In this process, GDP-mannose is converted to GDP-fucose through a three-step reaction, catalyzed by two enzymes, GDP-mannose 4,6-dehydratase (GMDS) and GDP-fucose synthetase (TSTA3, also known as FX) (Figure 1).[39] Alternatively, GDP-fucose can be synthesized via the diet/salvage pathway, directly from fucose.[40–43] Here, fucose is first phosphorylated by fucokinase (FCSK) to fucose 1-phosphate (Fuc-1P), which is further converted to GDP-fucose by guanylyltransferase (FPGT) (Figure 1).[44] In mid-1970s, isotope dilution studies showed that >90% of GDP-fucose in HeLa cells was supplied by the *de novo* pathway and <10% came from salvage.[42,43] However, it should be pointed out that while the study was meticulously performed, only a single fucose concentration was used. Preliminary studies suggest that fucose may make a more substantial contribution than the previous results showed.[45]

Fucose occurs in glycans as α-fucose and is released by α-fucosidase.[46] Fucose mutarotase converts it to β-fucose, and only the β-anomer is metabolized through the salvage pathway.[44] The enzyme is highly expressed in the liver, suggesting that it may be important for the fucose salvage, but little is known about this protein.[47]

Not much is known about fucose transport by human cells. In 1994, it was shown that it is a Na^+-independent mechanism, which occurs through an unidentified carrier system. Once transported into the cell, fucose is quickly incorporated into glycoproteins, and a large portion of these glycoproteins is rapidly secreted.[48] Incorporation requires the delivery of GDP-fucose into the Golgi apparatus and ER lumen. In humans, this transport process is mediated by either SLC35C1 or SLC35C2 (Figure 2). While both are Golgi-localized, the former is thought to be the major provider of Golgi GDP-fucose,[49] while the latter is used for Notch1 fucosylation, which occurs in the ER.[50] In contrast to *Drosophila*, no ER-localized GDP-fucose transporter has been identified in humans.[51]

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To date, several genetic disorders of fucosylation are known, including pathological variants in *SLC35C1* that lead to leukocyte adhesion deficiency II (LAD II), also known as SLC35C1-CDG.[52] In these individuals, fucosylation is perturbed due to the malfunction of SLC35C1 causing immunodeficiency and severe mental and growth retardation, in addition to its hallmark feature, an elevated level of circulating leukocytes.[49] In a few cases, supplementing the diet with fucose can normalize elevated leukocytes, by promoting the synthesis of Sialyl-Lewis X on neutrophils and therefore allowing their rolling on the endothelium.[53,54] How a small amount of fucose is such an effective treatment is poorly understood and requires further studies. One possibility is that responsive cells prefer the salvage pathway over the *de novo* pathway and are able to direct a larger amount of GDP-fucose to the Golgi. Presumably patients with some residual transporter activity might respond to a higher concentration of GDP-fucose, while others with more severe defects do not. No studies have been done to differentiate the responses of various mutations. Oral fucose therapy may also be helpful in FUK-CDG and FUT8-CDG. The former is caused by pathogenic variants in *FCSK*, which encodes fucose kinase,[55] and the latter is a result of biallelic mutations in *FUT8*,[56] which encodes the major fucosyltransferase required for adding core fucose to N-glycans.[57] No trials on these very few individuals have been performed.

The benefits of dietary fucose supplementation could have broader implications. Feeding fucose to mice or increasing the level of expression of *Fcsk* attenuated primary melanoma growth, increased the number of tumor-associated natural killer cells, and decreased the extent of distal metastasis of the tumor.[58] This appears to be due to the tumors' downregulation of *Fcsk*. Like mannose, fucose supplementation can affect the host microbiome. Mouse models have shown that fucose decreased levels of endotoxin-producing bacteria from the Desulfovibrionaceae family. The same study showed that fucose supplementation reduced the incidence of both obesity and fatty liver induced by a high-fat diet.[59] Interestingly, inhalation of a fucose and galactose solution was beneficial for individuals with cystic fibrosis because it reduced *Pseudomonas aeruginosa* levels in sputum and decreased the risk of bacterial infection.[60]

D-GALACTOSE

Galactose is a key source of energy for infants and a crucial component of glycoconjugates. It is also transferred across the plasma membrane using the GLUT transporters.[61] Its concentration in blood does not exceed 10 μM.[62] Most galactose is catabolized via the Leloir pathway (Figure 3).[63] Galactose can also be utilized in glycosylation through multiple entry points. Before incorporation into the glycans, galactokinase (GALK1) phosphorylates galactose to generate galactose 1-phosphate (Gal-1P).[64] The predominant pathway in liver converts Gal-1P to UDP-galactose using uridinyltransferase (GALT), which is an exchange reaction involving UDP-glucose, generating glucose 1-phosphate (Glc-1P).[65] In addition, UDP-glucose pyrophosphorylase (UGP), the same enzyme that converts Glc-1P to UDP-glucose, exhibits residual activity toward Gal-1P (Figure 1).[66,67] Alternatively, uridine diphosphate galactose pyrophosphorylase (EC 2.7.7.10) converts Gal-1P to UDP-galactose.[68,69] Another pathway producing UDP-galactose from UDP-glucose involves an NAD-dependent reaction catalyzed by UDP-galactose epimerase (GALE) (Figure 1).[70] The

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interplay and contribution of each pathway to glycan biosynthesis in different cell types have not been studied, and it may be difficult to extrapolate results from one type of cell to another.

UDP-galactose produced by the various pathways then serves as the substrate for galactosyltransferases located in the Golgi and ER.[71] UDP-galactose is transported into these organelles' lumen from the cytosol through the major UDP-galactose transporter (SLC35A2) (Figure 2).[72] Galactose is released from glycans as the β-anomer using β-galactosidase, but only the α-anomer can be activated from salvage. This epimerization is catalyzed by galactose mutarotase (GALM).[73] In 2019, studies showed that mutations in *GALM* are associated with galactosemia-like symptoms, suggesting the existence of a novel type (IV) galactosemia.[74,75]

Galactosemia is due to pathogenic variants in one of the genes involved in the Leloir pathway, namely, *GALK1*, *GALT*, or *GALE* (Figure 3).[76] In this metabolic disorder, even small amounts of galactose are very harmful as this sugar and its metabolites accumulate without being further catabolized.[77] As already mentioned, besides GALT and GALE, two other enzymes yield UDP-galactose. UGP (EC 2.7.7.9) exhibits very little activity toward Gal-1P. In the 1950s and 1960s, the existence of enzymatic activity catalyzing the reaction between UTP and Gal-1P was demonstrated in rat and human liver.[68,69] UDP-galactose pyrophosphorylase (EC 2.7.7.10) is poorly described in the literature. It is unclear whether the UDP-galactose pyrophosphorylase enzymatic activity was catalyzed by UGP. Because some symptoms of galactosemia (GALT) remain even on an essentially galactose free diet, it is possible that some cells may prefer the UDP-galactose pyrophosphorylase pathway for glycosylation, and galactose limitation may lead to underglycosylation in those cells. Defective glycosylation is a typical feature of individuals with galactosemia, but it only partially improves with treatment.[78–81] However, glycosylation defects and its improvement have not been the main focus of therapy.

Galactose supplementation successfully treated several different types of CDG. In a single individual with SLC35A2-CDG, galactose restored galactosylation and seemed to have a positive impact on the neurological symptoms,[82] although it is unclear if this improvement was solely due to galactose supplementation, as some patients improve their abnormal glycosylation over time. Because the cohort of diagnosed SLC35A2-CDG individuals is growing rapidly, conclusions about the benefit of galactose therapy can be examined in more patients.[83,84] In PGM1-CDG, pathogenic variants in *PGM1* ultimately result in decreased enzyme activity, which interconverts glucose 6-phosphate (Glc-6P) to Glc-1P, resulting in slightly abnormal glycogen metabolism and dramatically abnormal glycosylation.[85] The defect can be partially bypassed by galactose supplementation, because it contributes as an alternative source of Glc-1P and also increases the size of the cytosolic pools of UDP-glucose and UDP-galactose (Figure 1). This therapy is successfully applied in the growing number of PGM1-CDG individuals.[86] However, it does not normalize glycosylation completely, and transferrin profiles may remain abnormal.[62]

SLC39A8-CDG results from mutations in the membrane-localized manganese transporter. Dysfunction in SLC39A8 transport activity affects the availability of cellular Mn^{2+} and thus

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the enzymatic activity of manganese-dependent glycosyltransferases, most notably β-1,4-galactosyltransferase, leading to impaired galactosylation. Oral galactose supplementation normalizes glycosylation in SLC39A8-CDG individuals, as it presumably increases the UDP-galactose concentration, which becomes more available for the β-1,4-galactosyltransferase.[87] Glycosylation in individuals with TMEM165-CDG improves slightly with galactose therapy.[88] TMEM165 most likely supplies Ca^{2+} and Mn^{2+} to the Golgi apparatus in exchange for protons, making it important in glycosyltransferase function and pH homeostasis.[89] Probably the same mechanism underlies the successful galactose therapy of SLC39A8-CDG and TMEM165-CDG individuals. Studies of therapeutic applications of galactose will be expanded as part of a large multicenter U.S. consortium of physicians, scientists, and patients recently funded by the National Institutes of Health.

Besides congenital disorders of glycosylation, oral application of galactose is beneficial in individuals with idiopathic steroid-resistant nephrotic syndrome. This therapy decreases focal segmental glomerulosclerosis but fails to improve proteinuria.[90] As mentioned previously, inhalation of galactose and fucose decreases the chances of *P. aeruginosa* infections in individuals suffering from cystic fibrosis.[60] In addition, oral galactose treatment might be beneficial for people with sporadic Alzheimer's disease being an alternative for incretin-based therapy. In a rat model of Alzheimer-like pathology induced with streptozotocin, exposure for 2 months to oral galactose improved the learning and memory function of rats. It also prevented the progress and improved already developed cognitive deficiency in the early stage of the disease.[91,92]

In the 1960s, HepG2 hepatocellular carcinoma cells were shown to increase respiration rates when grown in galactose-containing medium to maintain the ATP level.[93] More recent studies showed that replacing glucose with galactose increases the sensitivity of HepG2 cells to mitochondrial toxicity,[94] which most likely occurs due to the enhancement of mitochondrial oxidative phosphorylation by galactose.[95] However, these studies use an unrealistic level of 10 mM galactose, which vastly exceeds the physiological plasma concentration (~10 μM) of this sugar.[62] Therefore, its practical value is limited.

N-ACETYL-D-GLUCOSAMINE

GlcNAc is found in all types of glycans. How GlcNAc enters the cell is currently unknown, but to be incorporated into glycans, it must first be activated to UDP-GlcNAc. Different pathways contribute to the *de novo* biosynthesis of UDP-GlcNAc (Figure 1). One is the hexosamine biosynthetic pathway, which requires glucose, glucosamine, acetyl-coenzyme A, and UTP.[96] Here, glucosamine:fructose-6-phosphate-amidotransferase (GFAT) catalyzes the rate-limiting step, resulting in the production of glucosamine 6-phosphate (GlcN-6P) from Fruc-6P.[97] Alternatively, glucosamine can be directly phosphorylated by hexokinase (HK), allowing GlcN-6P to enter the pathway.[98] Via the action of GNA1, an acetyl group is transferred from acetyl-coenzyme A to GlcN-6P to form GlcNAc 6-phosphate (GlcNAc-6P).[99] Phosphoglucomutase 3 (PGM3) isomerizes GlcNAc-6P to GlcNAc 1-phosphate (GlcNAc-1P),[100] and UDP-*N*-acetylhexosamine pyrophosphorylase (UAP1) catalyzes the production of UDP-GlcNAc.[101] It is important to point out that human UDP-galactose 4-epimerase (GALE) can also interconvert UDP-GlcNAc and UDP-N-acetylgalactosamine,

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creating an alternative source of the *de novo* UDP-GlcNAc synthesis as well as its utilization.[70] Its importance and contribution to various glycans are unknown.

Furthermore, UDP-GlcNAc can be provided by the salvage pathway (Figure 1).[102] GlcNAc, which is recycled from lysosomal degradation of oligosaccharides or acquired from dietary sources, is phosphorylated by GlcNAc kinase (NAGK) to GlcNAc-6P and is processed further by PGM3.[103] In addition to GlcNAc, NAGK is also able to phosphorylate *N*-acetylmannosamine (ManNAc).[104] GlcNAc-2-epimerase (RENBP) catalyzes the interconversion of GlcNAc and ManNAc, preferably isomerizing the former monosaccharide to the latter.[105] The relative contributions of all of these pathways in any cell type have not been studied.

Four UDP-GlcNAc transporters, SLC35A3,[106] SLC35A5,[107] SLC35B4,[108] and SLC35D2,[109] contribute to its transfer across the Golgi and ER membranes, where it is used by different glycosyltransferases in glycan synthesis (Figure 2). UDP-GlcNAc is the one of two nucleotide sugars utilized in the cytosol.[7] Nucleocytoplasmic modification of serine and threonine residues by O-GlcNAc serves as an important metabolic sensor.[110] In addition, the cytoplasmic pool of UDP-GlcNAc can be used for hyaluronan biosynthesis, which is the only glycosaminoglycan synthesized in the cytoplasm.[8]

PGM3-CDG is caused by pathogenic mutations in phosphoglucomutase 3, which interconverts GlcNAc-1P and GlcNAc-6P, disrupting the synthesis of UDP-GlcNAc and ultimately resulting in abnormal glycosylation.[100] When PGM3-CDG fibroblasts were grown in medium supplemented with GlcNAc, abnormal glycosylation improved.[111] This finding prompted a small trial in which PGM3-CDG individuals were supplemented with GlcNAc. Unfortunately, the treatment did not replicate the glycosylation improvements seen in the cell-based model, and the trial was stopped. In 2000, oral application of GlcNAc was adopted as a therapy in 12 children with chronic inflammatory bowel disease. This treatment resulted in histological improvement of rectal biopsy, significantly increasing epithelial and lamina propria glycosaminoglycans and intracellular GlcNAc.[112] However, there is no information about further clinical trials for this therapy. Rabbits with experimental osteoarthritis showed improvement, when injected with GlcNAc, suggesting chondroprotective and anti-inflammatory effects of this monosaccharide.[113] Glucosamine and GlcNAc treatment of mouse osteoblasts increased the rate of bone matrix deposition and decreased the rate of bone resorption modulating bone metabolism in osteoarthritis.[114] In 2017, it was demonstrated that dietary supplementation with GlcNAc may improve the type II collagen metabolism of articular cartilage in healthy individuals, not exhibiting any arthritic syndromes, altogether suggesting GlcNAc as a potential therapeutic compound in osteoarthritis.[115] In addition to GlcNAc, glucosamine was also in use for osteoarthritis, but the studies yielded conflicting results.[116–118] In a mouse model for multiple sclerosis, an increased level of N-glycan branching was observed when GlcNAc was added to drinking water.[119] This is an important observation because rare variants in *MGAT5*, which encodes GlcNAc transferase, were identified as being associated with the severity of multiple sclerosis.[120] Disrupted N-glycan branching is thought to promote autoimmune demyelination and neurodegeneration in multiple sclerosis; therefore, correcting this phenotype by simply supplementing with GlcNAc is an attractive therapeutic strategy.[119]

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Another rare disorder is caused by deficiency in NGLY1, which is responsible for deglycosylation of misfolded, retrotranslocated proteins prior their degradation in the cytosol.[121] Affected individuals have developmental delay, liver disease, neurological symptoms, and alacrima.[122] In *dNGLY1*-deficient fruit flies, GlcNAc supplementation partially rescued lethality and had a positive impact on the longevity of the flies, possibly by restoring UDP-GlcNAc levels, although this increase could not be demonstrated.[123]

SIALIC ACID

The term “sialic acids” refers to more than 50 unique variations of two parental compounds, *N*-acetylneuraminic acid (Neu5Ac) and *N*-glycolylneuraminic acid (Neu5Gc). These nine-carbon sugars are typically at the nonreducing ends of N- and O-glycans and glycolipids.[124] Neu5Ac is the most abundant sialic acid in humans and can be synthesized *de novo* form UDP-GlcNAc in four steps (Figure 1). First, the bifunctional enzyme UDP-GlcNAc-2-epimerase/*N*-acetylmannosamine kinase (GNE) converts UDP-GlcNAc to ManNAc and subsequently to ManNAc 6-phosphate (ManNAc-6P).[125] Next, Neu5Ac 9-phosphate (Neu5Ac-9P) is synthesized from ManNAc-6P by NANS. Finally, NANP removes the phosphate residue producing Neu5Ac.[126] Recently, studies using *NANP* knockout cells showed that its activity is not essential for the *de novo* production of sialic acid, indicating an alternative phosphatase activity, bypassing NANP.[127]

Because of an inactivating mutation in *CMAH*, humans are genetically unable to synthesize Neu5Gc, which is the major sialic acid in most mammals (Figure 1).[128] Surprisingly, small amounts of this sialic acid are still present in human cells, almost certainly of a dietary origin.[129] While different pathways contribute to their uptake, including pinocytosis,[130] both Neu5Ac and Neu5Gc can be taken up from plasma and incorporated directly into glycans.[130,131] Neu5Ac can be salvaged from hydrolyzed glycans following its release by neuraminidases[132,133] and transport from the lysosomes via Sialin, a transporter encoded by *SLC17A5*.[134] Pathogenic mutations in *SLC17A5* can cause either infantile sialic acid storage disorder (also known as sialuria) or an adult form of sialuria known as Salla disease.[135]

KDN is deaminated sialic acid, and as mentioned above, Man-6P serves as a substrate in its production (Figure 1).[26] In contrast to Neu5Ac and Neu5Gc, KDN and KDN glycoconjugates are abundant in only lower vertebrates and pathogenic bacteria and undetectable in mammals.[136] *De novo* synthesis of free KDN by human cells is enhanced in hypoxia.[137]

In contrast to the other nucleotide sugars, CMP-Neu5Ac is synthesized in the nucleus and is the only one activated with a monophosphonucleotide.[5] Once synthesized, CMP-sialic acid is transported into the Golgi apparatus via SLC35A1 (Figure 2),[138] where it serves as a substrate for different sialyltransferases. Only a few individuals with SLC35A1-CDG have been diagnosed, and it is unclear if either sialic acid or ManNAc could be used for monosaccharide therapy.[139–142] One report showed that fibroblasts derived from a single individual do not improve glycosylation in media supplemented with sialic acid or ManNAc.[141]

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Biallelic mutations in *NANS* encoding the Neu5Ac synthase led to elevated levels of ManNAc and resulted in hyposialylation of glycoproteins and glycolipids.[143] Using *nansa* knockdown zebrafish embryos, it was demonstrated that addition of sialic acid to the water partially improved their phenotype by rescuing skeletal development. Interestingly, the effect of Neu5Ac supplementation was time-dependent, being successful immediately after its injection into embryos but not beyond 24 h after the fertilization.[143] In 2018, an experimental theapy of orally supplemented Neu5Ac that aimed to treat NANS-deficient individuals was started.

N-ACETYL-D-MANNOSAMINE

As mentioned in the previous section, ManNAc is a hexosamine precursor on the biosynthetic pathway of sialic acids (Figure 1). It is synthesized by GNE from UDP-GlcNAc and converted further to ManNAc-6P.[125] Then NANS and NANP use it as a substrate for Neu5Ac production (Figure 1).[126] Sialic acids can be degraded to ManNAc and pyruvate in the reaction catalyzed by *N*-acetylneuraminate pyruvate lyase (NPL).[144] This enzyme prevents Neu5Ac from being recycled.

GNE myopathy, also known as hereditary inclusion body myopathy, is a rare genetic muscle disorder caused by pathogenic variants in *GNE*.[145] A GNE-deficient mouse model showed a 25% decrease in the level of membrane-bound sialic acids, mainly through the reduction of polysialic acids on NCAM and a shift of the transferrin isoelectric point toward basic pH.[146] Different disease models showed that ManNAc or mannosamine can bypass the defect. For example, ManNAc added to the cell culture medium of GNE-CDG fibroblasts restored the observed glycosphingolipid defect.[147] In a separate GNE-deficient mouse model, ManNAc, sialic acid, and mannosamine rescued both kidney and muscle hyposialylation but only ManNAc improved proteinuria.[148] However, uptake of Neu5Ac administered orally to rats was insufficiently utilized and excreted mainly as free Neu5Ac via the kidneys. This could explain why the clinical trials of an extended release formulation of sialic acid (Ace-ER) were unsuccessful and discontinued in 2017.[149] In contrast to Neu5Ac, ManNAc is still in clinical trials for GNE myopathy.

In addition to GNE myopathy, pathogenic variants of *GNE* also lead to sialuria.[150] Massive urinary excretion of unconjugated Neu5Ac[151] is due to *GNE* mutations at an allosteric site that normally binds CMP-sialic acid and downregulates GNE activity.[152,153] In this disorder, altered sialic acid metabolism also causes developmental delays and coarse facial features.[151]

Studies in 2018 showed that pathogenic variants of *NPL* affect heart and skeletal muscle function and development and a *npl* knockdown in zebrafish mimicked the human phenotype. Treatment of zebrafish embryos with ManNAc and GlcNAc efficiently improved somite morphology and muscle fiber arrangement. Neu5Ac could also rescue the knockdown zebrafish phenotype but to a lesser extent. Interestingly, ManNAc, but not GlcNAc or Neu5Ac, improved heart looping.[154]

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RIBITOL

Ribitol is a sugar alcohol formed by the reduction of ribose and was initially identified as a component of the cell wall in Gram-positive bacteria. Most of their cell surface polysaccharides consist of teichoic acids, composed of ManNAc, GlcNAc, and ribitol phosphate.[155] There was no precedent for ribitol in human glycans, but this dramatically changed in 2016 when two ribitol 5-phosphate (Rbo-5P) residues were found on α-dystroglycan.[156] In humans, ISPD activates ribitol by CTP to CDP-ribitol (Figure 1).[157] It serves as a substrate for FKRP and FKTN enzymes, critical for incorporation of Rbo-5P into the O-mannose-based glycan chains of α-dystrogycan.[158] Prior to this discovery, pathogenic variants of *SLC35A1*, encoding the CMP-sialic acid transporter, produced abnormal α-dystroglycan O-mannosylation, independent of sialic acid,[142] suggesting SLC35A1 as a strong candidate for a CDP-ribitol transporter (Figure 2).

Pathogenic variants of *FKRP* and *FKTN* result in multiple clinically defined muscular dystrophies, including Walker-Warburg syndrome, muscle-eye-brain disease, and a mild form of limb-girdle disease.[159,160] Recently, mouse models demonstrated that oral supplementation with ribitol enhances the synthesis of CDP-ribitol and Rbo-5P. Importantly, this therapy significantly increases the level of functional glycosylation of α-dystroglycan in skeletal and cardiac muscles, thereby improving the condition of *Fkrp* mutant mice.[161] This finding might be a breakthrough in the treatment of FKPR-related muscular dystrophy; however, studies have not yet shown whether ribitol can reverse muscle deterioration.

MODIFIED MONOSACCHARIDES AS A TOOL FOR STUDYING GLYCAN FUNCTION

Metabolic labeling of glycans with chemically modified monosaccharides ushers in a novel avenue for the study of glycosylation. When combined with click chemistry, modified monosaccharides are a powerful tool for improving our understanding of sugar utilization within the cell and organism. Azido sugars, such as ManNAz, GalNAz, GlcNAz, and 6-azidofucose, are broadly applied for the visualization of cellular glycans and for glycoproteomics.[162] They can be utilized both in cell cultures and in model organisms, permitting the study of biochemical pathways involving different monosaccharides. Other analogues, including peracetyl, alkynyl, or alkenyl sugars, are also in common use. They are developed as not only glycan imaging probes but also glycosylation inhibitors and remodelers.[163–166] Other applications of chemically modified monosaccharides include targeting toxins to tumor cells, inhibiting pathogen binding, or altering immune cell activity.[167]

Despite their broad application range, these monosaccharides must be cautiously utilized. As nonphysiological components, they may not fully mimic the behavior of naturally occurring sugars and might have secondary effects, which are hard to predict. For instance, 6-alkynyl fucose competitively inhibits *de novo* GDP-fucose synthesis via the FX enzyme.[166] In contrast, 7-alkynyl fucose serves as a highly sensitive probe that labels fucose in glycans.[165] Also, studies of zebrafish embryogenesis revealed that azide-modified fucose is inefficiently processed by the fucose salvage pathway enzymes.[168]

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CONCLUDING REMARKS

Most of the studies of glycan-associated monosaccharides were performed in the 1960s and 1970s, using the best methods, reagents, and biological materials of that era. Today, researchers consider these studies as the dogma of bedrock biochemistry. For many monosaccharides, their mechanism of transport, contribution to nucleotide sugar synthesis, and turnover rates have not been studied in any detail. Results can vary according to cell lines as some may preferentially use one pathway over the others. Many studies were conducted using cancer cells or cell lines lacking entire pathways (e.g., HCT116, CHO-Lec13, or Neuro-2A deficient in the *de novo* GDP-fucose synthesis). In most cases, data from model organisms are lacking. A few studies demonstrate cross-talk between different pathways and the importance of using physiological concentrations of monosaccharides, which vastly differ between monosaccharides.[23,30,45,153,169] Intermediate products of one pathway can produce allosteric inhibition of the enzymes in another pathway leading to the same product, as shown for GDP-fucose biosynthesis in the 1960s and 1970s when bacterial homologues of human enzymes were characterized.[170] In addition, the origin of the activated monosaccharide may affect its utilization or access to seemingly continuous pathways, as shown by the lack of entry of mannose-derived glucose into glycogen or its exit.[23] Most published studies do not differentiate between the salvage and dietary origin of monosaccharides (Table 1), and these two terms are often treated equivalently; here, we refer to only a few examples of this very common mistake.[171,172] It is also very hard to assess if the sugar was salvaged or secreted and taken back into the same cell or a different cell before its reutilization. Incorporation of monosaccharides through the diet and salvage should be considered as distinct pathways as was shown for mannose, where dietary mannose is either used directly for glycan synthesis or used as fuel but released mannose is secreted into the medium.[23] Salvage can be assessed by blocking lysosomal degradation of glycans or preventing the exit of the monosaccharide from the lysosome, as it was done for GlcNAc and sialic acid recycling.[102,133] Another way is to add specific inhibitors to block activation and reincorporation of the released sugar into glycans.

As monosaccharides become a popular therapy for an ever-growing number of congenital disorders of glycosylation (Table 1), we need to revise some of the previous findings on the basis of new technologies not available to researchers 45–50 years ago. These new approaches may help overcome past limitations. It is also important to keep in mind that monosaccharides might be toxic and, under some conditions, are dangerous. Galactose is harmful for individuals with galactosemia;[76] mannose exhibits toxicity in *Mpi* hypomorphic mice,[34] and D-galactosamine leads to liver alterations in rats that resemble human viral hepatitis.[173] Filling the gaps will allow us to better understand the mechanisms behind both the successes and failures of a therapy, potential toxicities, and therefore provide better help to the afflicted individuals treated with monosaccharides.

Finally, some companies market indigestible polysaccharides as sources of “essential glyconutrients” that have rightly been branded “shams”.[174,175] It is important for the scientific community to guard against misrepresentation of the facts, and that is best done by filling in those blanks with data. If we do not, marketers of sham glyconutrients will be happy to present “alternative facts” on our behalf.

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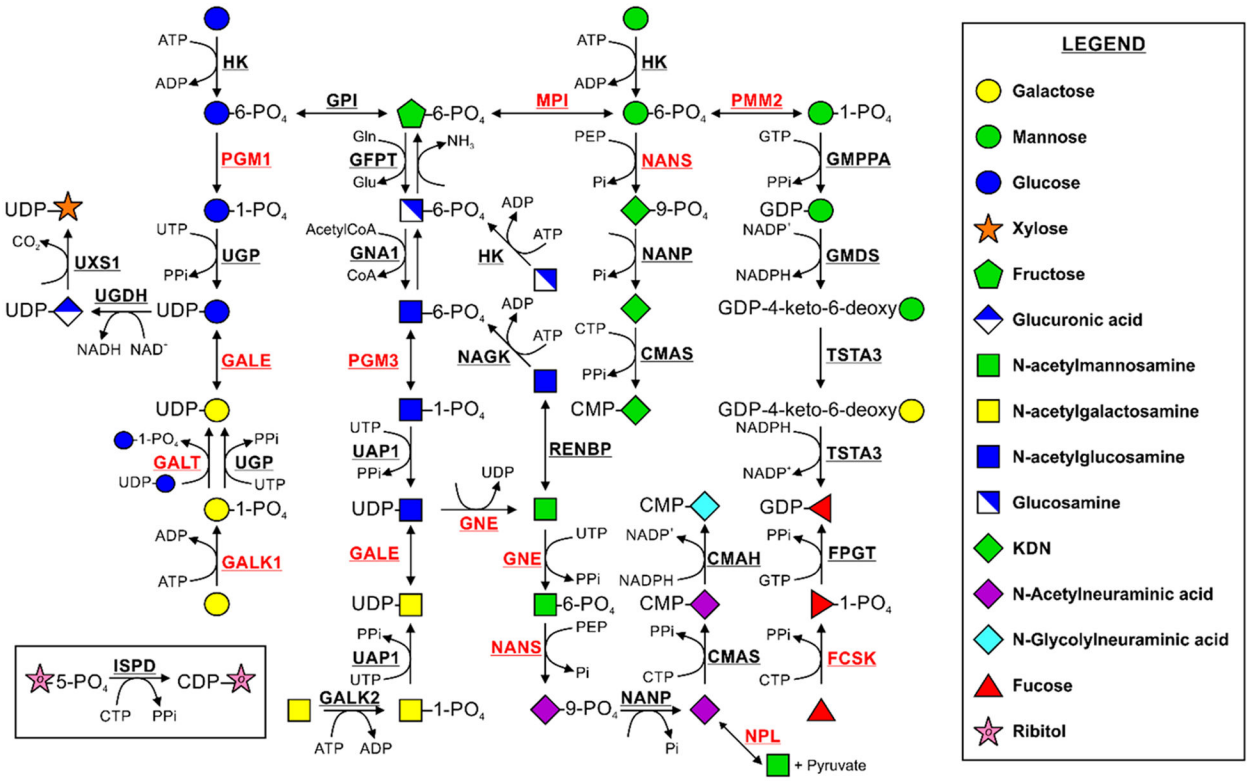


Figure 1.

Biosynthesis and interconversion of monosaccharides. Names of enzymes catalyzing each reaction are shown in bold. Those with pathogenic variants discussed in this review are shown in bold red. Adapted from *Essentials of Glycobiology*.[1]

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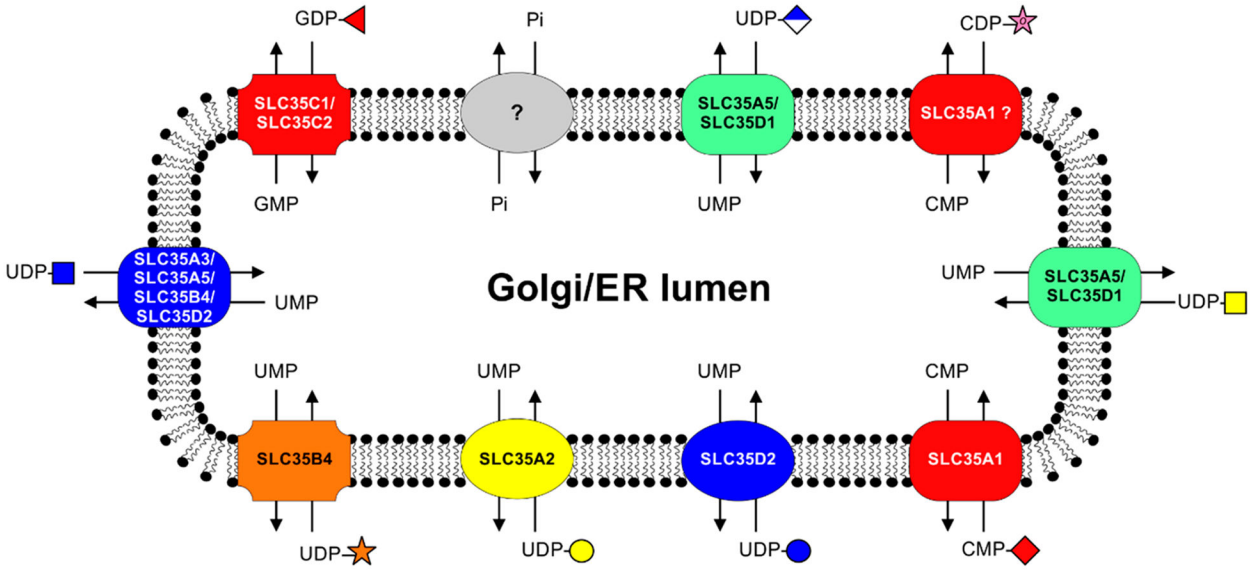


Figure 2.

Nucleotide sugar transporters delivering activated forms of monosaccharides into the Golgi and ER lumen in humans. The red diamond denotes sialic acid. Adapted from *Essentials of Glycobiology*.[1]

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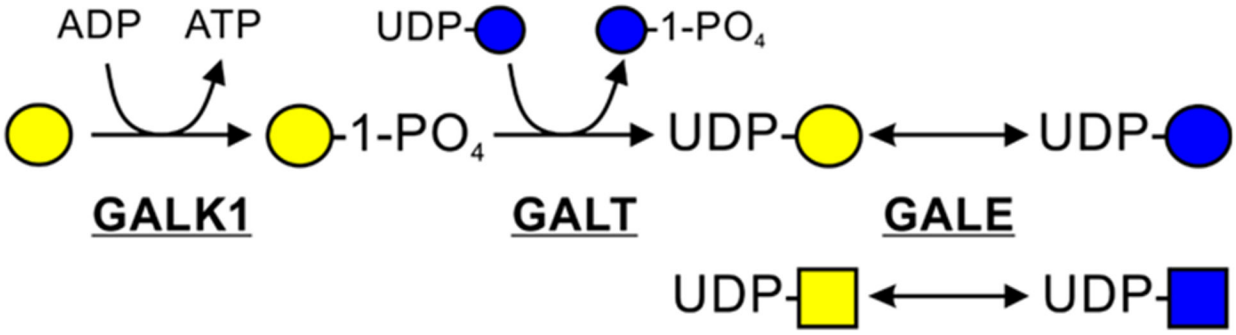


Figure 3.
Leloir pathway of galactose catabolism. Adapted from *Essentials of Glycobiology*.[1]

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Table 1.

Summary of the Most Important Information about the Monosaccharides Discussed Herein

Monosaccharide	Major pathways	Therapeutic applications	Comments
Mannose	-Glycosylation (not glycolipids) *_de novo_ synthesis from glucose[29] *from diet, activated by GMPP[23] *KDN synthesis[26] -Catabolized[23]	-Supplement for MPI-CDG[32] -Treatment and prophylactic for urinary tract infections[35]	-**TOXIC:** _Mpi_ hypomorphic mice[34] -**HARMFUL:** may lead to diarrhea[176] -Modulates host's microbiome[36] -Beneficial in obesity and cancer[36, 37] -**NOT** converted to glucose in glycogen[23] -**NOT** salvaged[23]
Fucose	-Glycosylation *_de novo_ synthesis from mannose and glucose[39] *salvaged[# 42, 43] or from diet[#,40] activated by FPGT -Oxidized to lactate[177]	-Supplement for fucosylation abnormalities due to impaired Golgi transport of GDP - fucose (LADII)[53, 54]	-Modulates host microbiome[59] -Beneficial in obesity and cancer[58] -Decreases bacterial infection in cystic fibrosis[60]
Galactose	-Glycosylation * synthesized _de novo_ from glucose[70] *salvaged[#] or form diet[#] activated by GALT[65] -Catabolized through the Leloir pathway [63] -Oxidized to CO_2[178]	Supplement for galactosylation abnormalities due to: -Decreased Golgi transport of UDP-galactose (SLC35A2-CDG)[82] -Insufficient isomerization of Glu-6P to Glc-lP (PGM1-CDG)[86] -Dysfunction in cellular transport of Mn^{2+} (SLC39A8-CDG)[87] -Deficiency in the Golgi transport of Mn^{2+} and Ca^{2+} (TMEM165-CDG)[88]	-**TOXIC:** in galactosemia[76] -**HARMFUL:** in lactose intolerance[179] -Decreases the risk of bacterial infection in cystic fibrosis[60] -Beneficial in idiopathic steroid-resistant nephrotic syndrome[90]
GlcNAc	-Glycosylation *synthesized _de novo_ in hexosamine biosynthetic pathway[96] or from UDP-GalNAc[70] * salvaged[102] or from diet,[112] activated by UAP1	-Supplement of glycosaminoglycans synthesis in chronic inflammatory bowel disease[112] -Joint pain and damage in osteoarthritis[113] -Impaired N-glycans branching in multiple sclerosis[119]	-GlcNAc as a therapy in GlcNAc-6P→ GlcNAc-1P isomerase activity deficiency (PGM3-CDG) failed clinical trials[111] -Unclear if it has a positive impact in osteoarthritis -In _dNGLY1_ knock-down fruit flies GlcNAc increased survival and longevity[123]
Sialic acid	-Glycosylation * synthesized _de novo_ from GlcNAc or ManNAc[125] *salvaged[## 133] or from diet[131], activated by CMAS -Catabolized to ManNAc and pyruvate[144]	-Defect in dephosphorylation of NeuNAc-9P mediated by NANS[143]	-Ineffective in impaired CMP-sialic acid transport (SLC35A1-CDG) fibroblasts[141] -Failed as a therapy for GNE myopathy[149]
ManNAc	-Glycosylation *precursor for sialic acid and GlcNAc[105, 125] -Product of sialic acid catabolism[144]	-GNE myopathy[180] -NPL defects with perturbed sialic acid catabolism in zebrafish[154]	-ManNAc therapy of NPL deficiency used in humans -Not found in human glycans
Ribitol	-Glycosylation: *incorporated as Rbo-5P into α-dystroglycan[156]	Pathogenic variants of _FKRP_, adding Rbo-5P to α-dystroglycan, resulting in: -muscular dystrophy -Walker-Warburg syndrome -muscle-eye-brain disease -mild form of limb-girdle disease[161]	-Ribitol therapy untested in humans -Defect in α-dystroglycan O-mannosylation identified in SLC35A1-CDG individual[142] -High ribitol concentrations found in ribose-5-phosphate isomerase deficiency (pentose-phosphate-pathway defect)[181]

[a]The contribution and the difference between diet and salvage were not determined.

[b] Salvage from glycoproteins was not studied.

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